

Practical week 7:

```
PS C:\Users\uday\Downloads\Practical 7(DB)\nodememo\user-service> npm install express mongoose dotenv

added 87 packages, and audited 88 packages in 16s

30 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

```
✓ Practical 7(DB)
  ✓ nodememo\user-service
    > controllers
    > node_modules
    > routes
  JS app.js
  JS db.js
  {} package-lock.json
  {} package.json
```

Explorer

Practical 7(DB)

- nodememo\user-service
 - controllers
 - node_modules
 - routes
 - JS app.js
 - JS db.js
 - package-lock.json
 - package.json
- Outline
- Timeline

```
1  const mongoose = require("mongoose");
2
3  const MONGO_URI =
4    process.env.MONGO_URI ||
5    "mongodb://127.0.0.1:27017/digital_library";
6
7  mongoose.connect(MONGO_URI)
8    .then(() => console.log("MongoDB connected"))
9    .catch((err) => console.error("MongoDB connection error:", err));
10
11 const userSchema = new mongoose.Schema({
12   username: {
13     type: String,
14     required: true
15   },
16   email: {
17     type: String,
18     required: true
19   },
20   profile: {
21     type: mongoose.Schema.Types.Mixed,
22     default: {}
23   }
24 });
25
26 const activitySchema = new mongoose.Schema({
27   user_id: {
28     type: mongoose.Schema.Types.ObjectId,
29     required: true,
30     ref: "User"
31   },
32   action: {
33     type: String,
34     required: true
35   },
36
```

Practical 7(DB)

nodememo\user-service

- controllers
 - JS userController.js
- node_modules
- routes
- JS app.js
- JS db.js
- package-lock.json
- package.json

```
1  const { User } = require("../db");
2
3  // POST - Create User
4  const createUser = async (req, res) => {
5    try {
6      const { username, email, profile } = req.body;
7
8      const user = await User.create({
9        username,
10       email,
11       profile
12     });
13
14     res.status(201).json({
15       message: "User created successfully",
16       user
17     });
18   } catch (error) {
19     res.status(500).json({
20       message: "Error creating user",
21       error: error.message
22     });
23   }
24 }
```

```
JS userController.js  JS activityController.js  JS userRoutes.js X
nodememo > user-service > routes > JS userRoutes.js > ...
1  const express = require("express");
2  const router = express.Router();
3
4  const {
5    createUser,
6    getUser
7  } = require("../controllers/userController");
8
9  router.post("/users", createUser);
10 router.get("/users/:id", getUser);
11
12 module.exports = router;
```

Suggestion [3,General]: The command DB was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default. If you trust this command, instead type: ".\DB". See "get-help about_Command_Precedence" for more details.

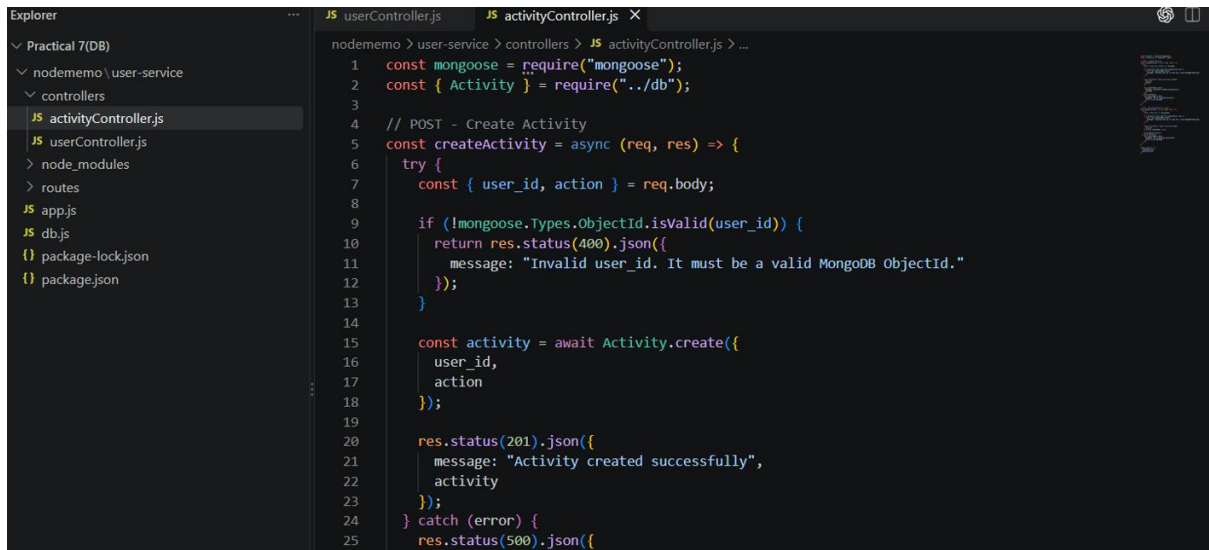
```
PS C:\Users\uday\Downloads\Practical 7(DB)\nodememo\user-service> node app.js
User service running on port 5001
MongoDB connected
█
```

```
PS C:\Users\uday\Downloads\Practical 7(DB)\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/users" -Method POST -ContentType "application/json" -Body '{"username":"Divya","email":"divya@gmail.com","profile":{}}'
```

message	user
-----	----
User created successfully @{username=Divya; email=divya@gmail.com; _id=6abd39990d9bbd8fef6fd7b9; __v=0}	

```
PS C:\Users\uday\Downloads\Practical 7(DB)\nodememo\user-service> █
```

```
Pretty-print ☒
{
  "_id": "6abd39990d9bbd8fef6fd7b9",
  "username": "Divya",
  "email": "divya@gmail.com",
  "__v": 0
}
```

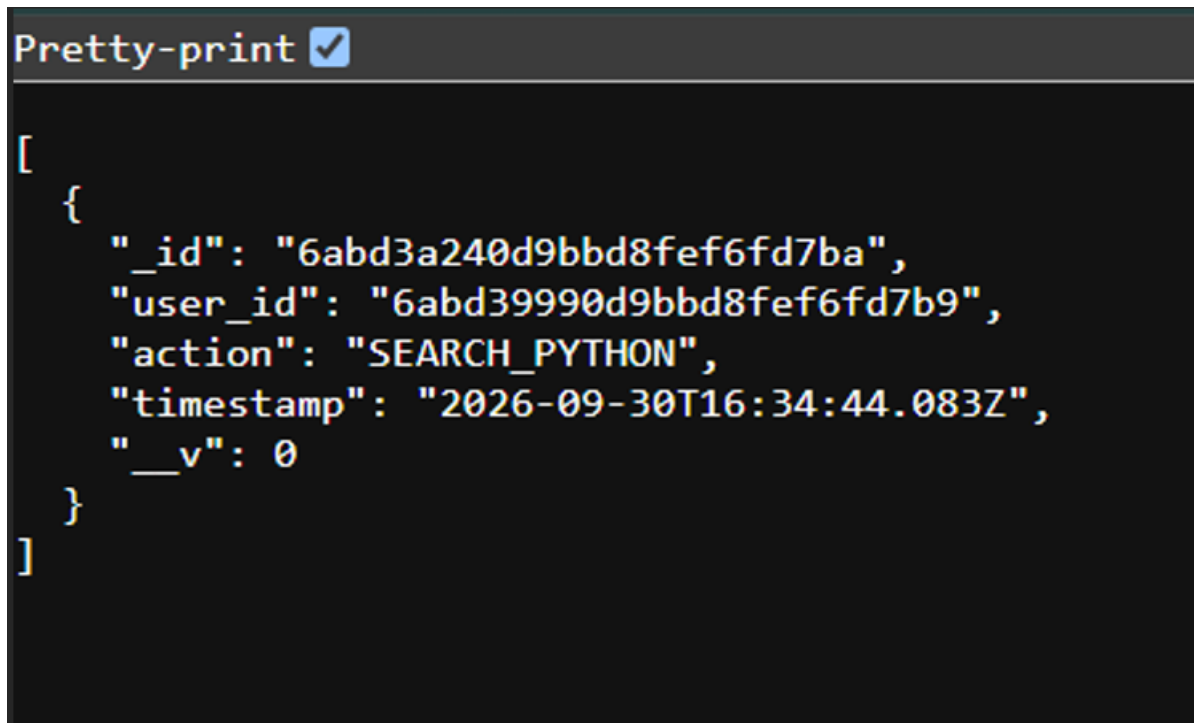


```
1 const mongoose = require("mongoose");
2 const { Activity } = require("../db");
3
4 // POST - Create Activity
5 const createActivity = async (req, res) => {
6   try {
7     const { user_id, action } = req.body;
8
9     if (!mongoose.Types.ObjectId.isValid(user_id)) {
10       return res.status(400).json({
11         message: "Invalid user_id. It must be a valid MongoDB ObjectId."
12       });
13     }
14
15     const activity = await Activity.create({
16       user_id,
17       action
18     });
19
20     res.status(201).json({
21       message: "Activity created successfully",
22       activity
23     });
24   } catch (error) {
25     res.status(500).json({
```

```
PS C:\Users\uday\Downloads\Practical 7(DB)\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/activities" -Method POST -ContentType "application/json" -Body '{"user_id":"6abd39990d9bbd8fef6fd7b9","action":"SEARCH_PYTHON"}'
```

```
message          activity
-----
Activity created successfully @{user_id=6abd39990d9bbd8fef6fd7b9; action=SEARCH_PYTHON; _id=6abd3a240d...
```

```
PS C:\Users\uday\Downloads\Practical 7(DB)\nodememo\user-service>
```



```
[
  {
    "_id": "6abd3a240d9bbd8fef6fd7ba",
    "user_id": "6abd39990d9bbd8fef6fd7b9",
    "action": "SEARCH_PYTHON",
    "timestamp": "2026-09-30T16:34:44.083Z",
    "__v": 0
  }
]
```

Verifying in mongo db atlas



Type a query: { field: 'value' } or [Generate query](#) ✨

ADD DATA ▾

BULK ▾

EXPORT CODE

```
_id: ObjectId('6abd39990d9bbd8fef6fd7b9')
username: "Divya"
email: "divya@gmail.com"
__v: 0
```

```
_id: ObjectId('6abd3a240d9bbd8fef6fd7ba')
user_id: ObjectId('6abd39990d9bbd8fef6fd7b9')
action: "SEARCH_PYTHON"
timestamp: ISODate('2026-09-30T16:34:44.083+00:00') September 30, 2026 at 4:34:44 PM UTC
__v: 0
```